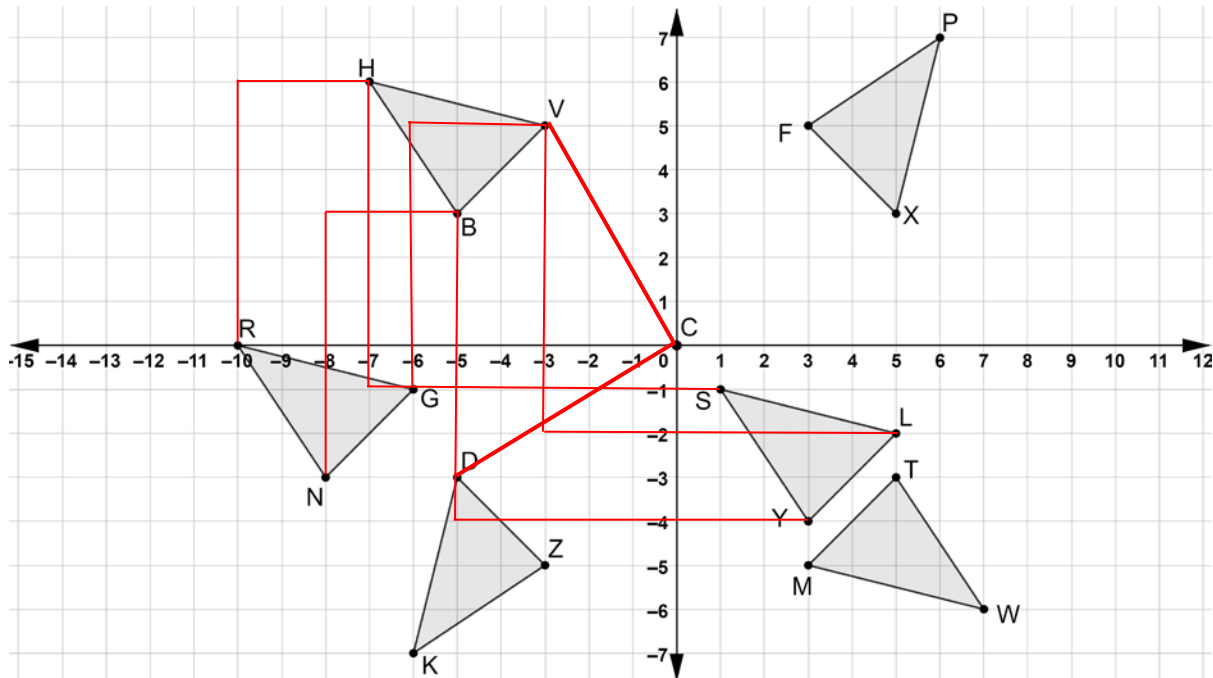


Geometry
Unit 4
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

Six triangles are shown on the coordinate grid.



1. Draw the mapping that shows the translation if $\triangle SYL$ is the preimage and $\triangle HBV$ is the image.
2. Draw the mapping that shows the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.
3. Describe the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.
 $\triangle HBV$ is shifted left 3 units and down 6 units.

4. Complete the statements.

The x -values of $\triangle RNG$, the image, can be found by

☐ adding
☒ subtracting

11 unit(s) to the x -values of $\triangle SYL$, the preimage. The y -values

of $\triangle RNG$, the image, can be found by

☒ adding
☐ subtracting

1 unit(s)

to the y -values of $\triangle SYL$, the preimage.

5. On the coordinate grid draw angle DCV .
6. The algebraic description of the rotation of $\triangle KZD$, the preimage, onto $\triangle HBV$ is the image, is $(x,y) \rightarrow (y,-x)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.

90° clockwise rotation

$K: (-6, -7)$	$H: (-7, 6)$
$Z: (-3, -5)$	$B: (-5, 3)$
$D: (-5, -3)$	$V: (-3, 5)$

7. The algebraic description of the rotation of $\triangle TWM$, the preimage, onto $\triangle BHV$ is the image, is $(x,y) \rightarrow (-x,-y)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.

180° rotation

$T: (5, -3)$	$B: (-5, 3)$
$W: (7, -6)$	$H: (-7, 6)$
$M: (3, -5)$	$V: (-3, 5)$

8. Describe how the measure of $\angle BCF$ can be used to determine the degrees of rotation where $\triangle PFX$ is the preimage and $\triangle HBV$ is the image.

$\angle BCF$ is a 90° angle. It shows that $\triangle PFX$ is rotated 90° counterclockwise.

9. Describe how the coordinates of the vertices of $\triangle PFX$, the preimage, and the coordinates of $\triangle HBV$, the image can be used to determine the degrees of rotation.

The y -coordinate is multiplied by -1 , then the x and y -values switch.

10. Describe how the orientation of the triangles helps to determine which transformations were a translation and which transformations were a rotation.

Translations do not change the orientation of a figure, while rotations do change the orientation.